CLUBDASH MASTER SYSTEM ARCHITECTURE & OPERATIONAL BLUEPRINT

Document Status: Master Asset / Single Source of Truth **System Version:** 1.2 (End-to-End B2B2C Ecosystem)

This document serves as the absolute foundational blueprint for the Clubdash platform. It dictates the engineering infrastructure, database schemas, business rules, and user experience (UX) flows. Developers and product managers must adhere to these specifications to maintain structural fidelity and ensure the ecosystem scales seamlessly.

1. Ecosystem Topology & Hardware Matrix

The Clubdash ecosystem decouples physical real estate ownership from community programming. The system serves three distinct actors operating on specific hardware environments to eliminate operational friction.

Actor Archetype	Role / Objective	Primary Software Interface	Access Status
Facility Owner/Staff	Real Estate / Landlord. Maximizes inventory utilization and B2B rentals.	**Facility Web OS** & **Quickdash** (Desktop/Tablet)	[LOCKED] Verified Owners
Organizer / Coach	Promoter / Talent. Manages rosters, builds funnels, sources venues.	**Clubdash Pro App** (Mobile-First iOS/Android)	[OPEN] Self-Serve
Player / Consumer	Consumer. Books courts, buys event tickets, consumes media.	**Sodia App** (Native Mobile Ecosystem)	[OPEN] Public

2. Engineering Infrastructure: The Agnostic Core

(Refer to core mechanics: Spatial Sub-Division, Multi-Sport Tagging, and the JSON-based Universal Bracket Engine).

3. Inventory Rules & Time Mechanics

(Refer to dynamic time-blocks and automated cross-sport changeover buffers).

4. The Dual-Membership Identity Matrix

(Refer to Facility Tier 0 Base Waivers, Facility Pro Tiers, and Organizer VIP limits).

5. B2B Marketplace & Financial Routing

(Refer to Digital Affiliation Handshakes and Stripe Connect Split Ledgers).

6.4 Sodia App (The Consumer Harness)

(Refer to Discovery Feed, POS Intercepts, and the Digital Wallet QR).

6.5 Staff Identity & DTR (The Accountability Engine)

- **Mobile Selfie Clock-In:** Staff clock in via personal Sodia app (geofenced, timestamped selfie).
- **The Float Gate & Blind Count:** Staff must enter a PIN and Opening Float on the Quickdash tablet to handle cash. Shift end requires a blind cash count to secure the ledger.

8. Business Intelligence & Revenue Protection

To ensure Facility Owners rely on Clubdash as their ultimate business engine, the following operational and analytics tools are hardcoded into the Web OS and Quickdash infrastructure (GymDash Integrations).

8.1 The Global Time-Frame Metrics Dash

The Facility Web OS features a master dropdown (`Daily, Weekly, Monthly, Quarterly, Annual, All-Time`). Toggling this instantly recalculates the entire command center:

- **The Growth Equation:** Visualizes total facility utilization, B2B vs. B2C revenue splits, and month-over-month growth.
- **The Platform ROI Card:** Explicitly isolates app-sourced consumer bookings and B2B Organizer payouts to mathematically prove Clubdash's value to the landlord.

8.2 Yield Management & The Hourly Heatmap

- **Dead-Hour Detection:** The system tracks historical court utilization by hour and weekday.
- **Flash Promos:** When a "dead hour" is identified (e.g., Tuesday at 11:00 AM), the Web OS surfaces an action button to draft an "Off-Peak SKU." The owner can publish a discounted court rate instantly to the Sodia consumer feed to monetize empty real estate.

8.3 B2B Predictive Attrition CRM

Organizers churning costs facilities thousands in lost recurring revenue.

- **The Dormant Queue:** If a verified Affiliate Organizer hasn't booked space in 30 days, they enter the "At-Risk/Dormant" list.
- **One-Tap Win-Back:** The Facility Owner can tap a single button to push a targeted wholesale discount directly into the Organizer's Clubdash Pro Inbox to secure their next tournament.

8.4 Squatter Detection (Gate-vs-Booking Gap)

- **The Alert Logic:** The system cross-references the Quickdash entry scans against the live court capacity limits.
- **Action:** If Court 1 is booked for 10 people, but Quickdash records 14 valid Tier 0 scans associated with that booking, the system flashes a "Capacity Breach" alert to the staff's mobile device, prompting them to enforce an upsell or perform a floor walk-through.